



DRIFT

the early-warning court for AI quality

Watches your AI in production, puts every suspicion of degradation on [trial](#), and hands you a [countdown](#) — hours before quality becomes unacceptable.

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github.com/Dan16ssd/Drift-ForAMDHACKATHON

THE PROBLEM

AI quality doesn't fail loudly. It drifts.

A retrieval index goes quietly stale. A model update shifts behavior. Every individual response still looks *fine* — so thresholds stay green while quality slides for **days**.

Your observability stack logged everything and told you nothing. You find out from **customers**.

And the tools that do alert? Teams mute them within weeks — because they can't tell degradation from a traffic shift.

3 days

typical silent-regression window — 10k conv/day bot → 30,000 bad customer experiences

\$3–8

per ticket escalated to human agents, plus churn you can never attribute

0

monitoring vendors that publish their own false-alarm rate

THE SOLUTION

DRIFT — the early-warning court for AI quality

production AI stream → **SENSOR** · scores every response → **LEDGER** · append-only, self-grading →

THE COURT · every alarm cross-examined → **COUNTDOWN** · hours-to-floor + cause

LLMs argue and explain.

Deterministic code measures and extrapolates. No model ever produces a number the system acts on — regression, changepoints, confounder stats are all scipy.

Uncertainty buys evidence, not silence.

A suspicious-but-unproven trend doesn't page a human and doesn't get dismissed — WATCH tightens the sampling rate until the debate can be settled.

The system grades itself.

Every alert's outcome is backfilled when the predicted crossing does (or doesn't) happen. Alert precision is the headline metric on the dashboard.

No alert reaches a human until it survives cross-examination

Prosecutor

Argues the decline is real — armed with slope, R^2 , p-value and a changepoint the **code** computed.

Defense

Attacks with every innocent explanation: traffic-mix shift, one outlier user, time-of-day, scorer noise, sample size.

Judge

Rules DISMISS / WATCH / ALERT on a fixed standard of proof, citing ledger rows. Judgment is auditable.

CI machine-checks the court's judgment on every commit

Fixture stream	What's planted	Required verdict
stable	noise only	no ALERT
drift	retrieval decay on a known schedule	ALERT + countdown, outcome confirmed
confounder	traffic mix shifts to hard topics — aggregate falls, nothing broken	no ALERT — the case that defeats threshold monitors

THE PRODUCT MOMENT

A countdown, not a red dot

*"Quality crosses your floor in ~3.2–5.5 hours at the current rate;
probable cause: **retrieval decay**."*

hours

of warning before the floor — with a confidence range, from pure regression math

a cause

named from the sensor channel that's actually trending (retrieval, hedging, truncation...)

honesty

when the fit is too weak, DRIFT refuses to guess — a confident number from a bad fit is theater

Then the receipt: the predicted window is drawn over what actually happened. In our planted-drift demo, **the real crossing landed inside the predicted window** — and the alert's outcome was backfilled automatically. Precision: 1/1, shown live on the dashboard.

Incumbents describe the past. DRIFT forecasts — and shows its error rate.

	LangSmith	Arize	Braintrust	DRIFT
Logs, traces, dashboards	✓	✓	✓	✓
Adversarial verification of every alarm	—	—	—	✓ prosecutor / defense / judge
Time-to-unacceptable forecast	—	—	—	✓ deterministic countdown
Self-reported alert precision	—	—	—	✓ outcome backfill

Who pays

Every team whose AI talks to customers — support, onboarding, internal copilots. Budget already exists under "AI reliability"; silent-regression pain is hitting them for the first time.

The wedge

A three-day silent regression becomes a two-hour incident with a named cause. That is the difference between churn and a config rollback.

Our moat is partly a hardware-cost artifact — and we say so

Scoring **every** response continuously, plus two extra reasoning passes per suspicion, is exactly the workload **per-token API pricing punishes**.

On one MI300X, **192 GB HBM** holds the high-volume sensing model *and* the court's judge simultaneously — flat cost, no per-token meter running against your alarm system.

Adversarial verification is the feature incumbents can't afford to build on API economics.

OpenAI-compatible by construction

Same pipeline points at vLLM/ROCm in a customer VPC — endpoint is 3 env vars, per-seat splittable.

One-command onboarding

```
./scripts/go_live_amd.sh <url> <key>
```

catalog probe → cast check → calibration go/no-go → live replay.

Concurrent sensing

Order-preserving worker pool saturates the GPU; verdicts byte-identical to sequential.

BUILT, VERIFIED, LIVE

Everything you saw runs today

27

tests green incl. the three
fixture-verdict assertions,
public CI

300

responses scored live on real
models — correct verdicts, zero
errors

0.846

scorer-label Pearson r in the
live calibration spike (repeat std
0.0094)

1/1

alert precision, outcome-
backfilled — graded against
planted ground truth

AI quality doesn't fail loudly. Now it doesn't have to fail silently either.

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FastAPI · React · SQLAlchemy · SciPy · vLLM/ROCm-ready

DRIFT — verdict first, alert second